Interactive 3D Solar PV System Simulator for Irradiance Estimation and Performance Analysis

Anjesh Singh (107123014), Karan Singh Bisht (107123054), Vedant Dorlikar (107123133)
Department of Electrical and Electronics Engineering
National Institute of Technology, Tiruchirappalli
Tiruchirappalli, Tamil Nadu, India
{107123014, 107123054, 107123133}@nitt.edu

Abstract—This paper presents a course-aligned, browser-based, interactive three-dimensional photovoltaic (PV) system simulator that unifies solar geometry, plane-of-array (POA) irradiance estimation, single-diode inspired PV electrical modelling, partial shading with bypass diode behaviour, and maximum power point tracking (MPPT) in a single teaching and analysis tool. The simulator is implemented as a single-page web application using React, TypeScript, Three.js, Recharts and Zustand, and runs entirely in the browser with no backend dependency. A dedicated NIT Tiruchirappalli view renders 308 real OpenStreetMap building footprints extruded over an Esri World Imagery satellite tile of the campus, and seeds the irradiance sliders from the NASA POWER reanalysis at 10.759 °N, 78.817 °E, where the annual mean global horizontal irradiance (GHI) is 5.55 kWh/m^2/day and the annual mean air temperature is 27.6 °C. Computed results reproduce the expected textbook behaviour: increasing irradiance raises current approximately linearly, rising cell temperature reduces open-circuit voltage, bypass diodes recover most of the mismatch-induced power loss under partial shading, and a Perturb & Observe MPPT loop converges on the maximum power point within fewer than twenty perturbations. A rooftop PV potential study for the NIT Trichy campus estimates approximately 15 MW$_\mathrm{p}$ of usable roof-mounted PV capacity assuming 60% usable roof fraction and a 180 W/m^2 module-density.

Index Terms—solar photovoltaics, solar geometry, irradiance estimation, single-diode model, partial shading, bypass diode, maximum power point tracking, educational simulator, OpenStreetMap, NASA POWER, NIT Tiruchirappalli

I. Introduction

The EE-F70 Wind and Solar Systems syllabus at NIT Tiruchirappalli covers solar geometry, irradiance and irradiation, photovoltaic cell and module characteristics, fill factor, single-diode modelling, temperature and irradiance effects, partial shading with blocking and bypass diodes, and maximum power point tracking. Existing open-source tools tend to address only one of these layers at a time: pvlib-python [1] models the electrical and irradiance side in Python without a 3D scene, while open-pv/simshady [2] focuses on GPU-accelerated 3D shading without a full PV electrical model. A student who wants to see, in one place, how moving the sun, tilting a panel, adding a shading obstacle, changing temperature, enabling a bypass diode, or running an MPPT controller each affects the same I–V curve, must today stitch together several tools.

This project addresses that gap by building an integrated, browser-based simulator that:

1) renders a live 3D scene with a tilted PV panel, a sun driven by date/time/latitude, optional shading obstacles, and real-time cast shadows;
2) computes standard solar angles and isotropic-sky POA irradiance, including daily integration for energy yield;
3) solves a single-diode inspired I–V curve with temperature and irradiance coefficients, reports I_sc, V_oc, I_mp, V_mp, P_max and fill factor, and combines three substring curves for partial shading scenarios;
4) animates two MPPT algorithms (Perturb & Observe and Incremental Conductance) against the live I–V curve;
5) grounds the abstract textbook exercise in a real location by rendering a 3D reconstruction of the NIT Trichy campus from OpenStreetMap footprints sitting on Esri World Imagery, with NASA POWER climate import at one click.

The simulator, its full source code, bundled OSM/NASA data, and this report are available at https://github.com/KaranSinghBisht/solar-pv-simulator.

II. System Overview

Figure 1 sketches the module graph. User-facing controls and presets feed a single Zustand state store. A chain of pure TypeScript modules — solarGeometry, irradiance, pvModel, shading, mppt and energy — turns that state into a full simulation result on every render. Seven React views consume the result and present different educational slices.

A. Implementation Stack

The app is a Vite-bundled React 18 + TypeScript 5 single-page application. Three.js draws both the panel scene and the interactive earth globe with custom shaders for the atmospheric glow and the sky gradient. Recharts renders the I–V, P–V and daily-irradiance plots. Zustand hosts the ∼30-field simulation state. External datasets shipped in public/data/ include the NASA POWER monthly climatology, 365 days of 2023 daily data, 308 OSM building footprints, 784 road segments, amenities

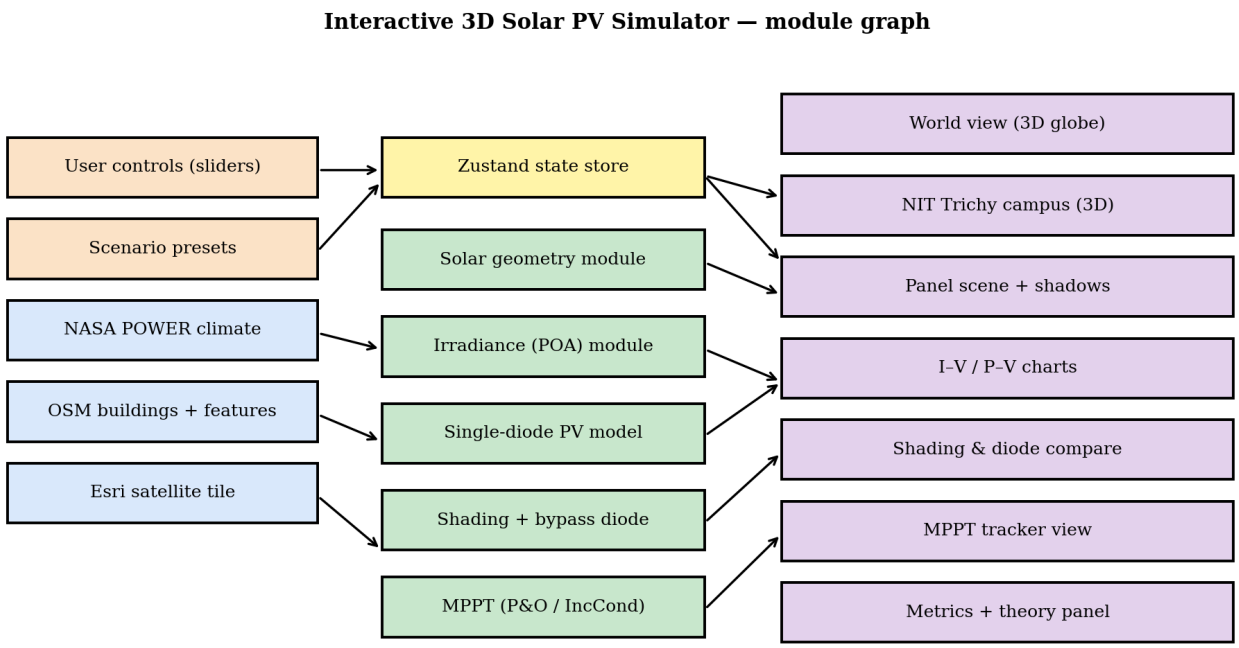

Fig. 1. Module and data-flow graph of the simulator. Orange blocks are user inputs, blue blocks are external datasets (NASA POWER, OpenStreetMap, Esri imagery), yellow is shared state, green is pure TypeScript physics, and purple are the React views.

and water towers, and a 2048×3091 Esri World Imagery tile of the campus.

B. Target Users

The tool is designed for three audiences identified in the course spec: faculty/examiners who need a technically sound academic simulator, students who want a tactile way to connect equations to physical intuition, and demo viewers who expect a visually engaging interface. All three are served by the same build; the left-hand control panel exposes every numerical knob, the tab bar organises the teaching modules in syllabus order.

III. Solar Geometry Module

A. Formulae

The day-number declination uses the Cooper approximation,

$$\delta = 23.45^\circ \sin\left(\frac{360(284+n)}{365}\right), \tag{1}$$

with hour angle

$$\omega = 15^\circ(t_{\text{solar}} - 12), \tag{2}$$

zenith

$$\cos\theta_z = \sin\phi\sin\delta + \cos\phi\cos\delta\cos\omega, \tag{3}$$

altitude $\alpha_s = 90^\circ - \theta_z$, and solar azimuth computed via arccos with the hour-angle sign convention. For the incidence angle on a tilted surface the implementation avoids the long closed-form expression by building two unit vectors in world coordinates,

$$\hat{\mathbf{s}} = (\cos\alpha_s \sin\gamma_s,\ \sin\alpha_s,\ \cos\alpha_s\cos\gamma_s), \tag{4}$$

$$\hat{\mathbf{n}} = (\sin\beta\sin\gamma,\ \cos\beta,\ \sin\beta\cos\gamma), \tag{5}$$

and computing

$$\cos\theta_i = \hat{\mathbf{s}}\cdot\hat{\mathbf{n}}. \tag{6}$$

This single dot product drives both the irradiance model and the 3D scene light direction.

B. Behaviour at NIT Trichy

Figure 2 plots the altitude and azimuth curves at 10.759 °N for four representative days. The near-equatorial latitude gives a peak altitude of roughly 80° at summer solstice noon and the azimuth symmetry around solar noon that the textbooks predict for tropical latitudes.

IV. Irradiance Estimation Module

The simulator distinguishes between instantaneous irradiance (W/m^2) and time-integrated irradiation (Wh/m^2 or kWh/m^2/day). POA irradiance on a tilted plane is decomposed using the standard isotropic-sky model:

$$G_{\text{beam}} = \text{DNI}\cdot\max(0,\cos\theta_i)\cdot f_{\text{SF}}, \tag{7}$$

$$G_{\text{diffuse}} = \text{DHI}\cdot\tfrac{1+\cos\beta}{2}, \tag{8}$$

$$G_{\text{ground}} = \rho\cdot\text{GHI}\cdot\tfrac{1-\cos\beta}{2}, \tag{9}$$

$$G_{\text{POA}} = G_{\text{beam}} + G_{\text{diffuse}} + G_{\text{ground}}. \tag{10}$$

Here $f_{\text{SF}} \in [0,1]$ is the shading fraction from the 3D scene or user-configured substring multipliers. Daily irradiation

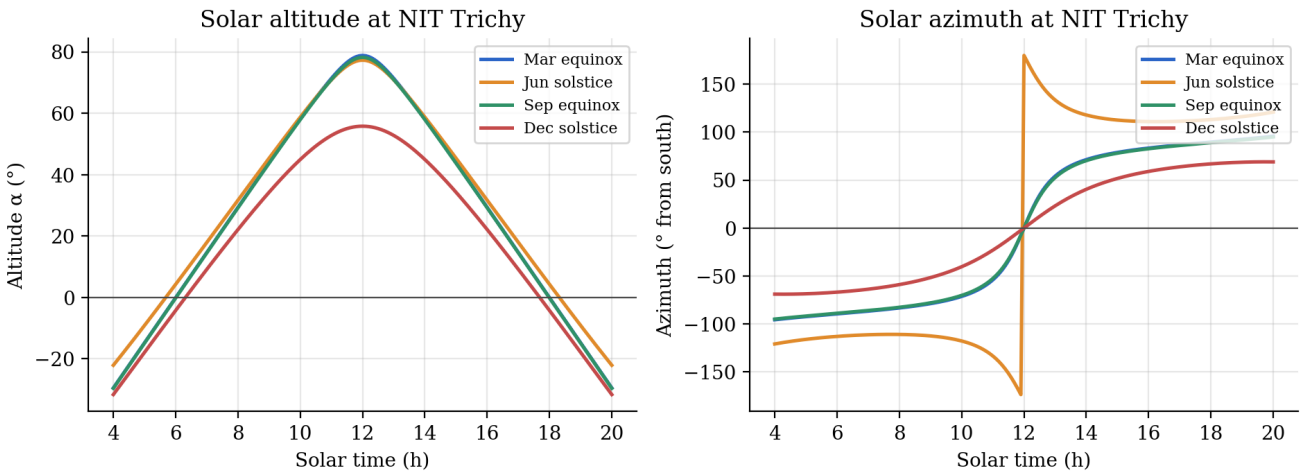

Fig. 2. Computed solar altitude and azimuth at NIT Trichy for four representative days of the year.

is obtained by integrating $G_{\mathrm{POA}}(t)$ every 0.5 h with a $\sin\alpha_s$ daylight envelope. The shading-factor model is kept deliberately simple and educational; for v1 the isotropic diffuse sky is used instead of more complex Perez/Hay-Davies transpositions.

Figure 3 shows the computed POA profile at NIT Trichy on the two solstices for two panel tilts. At June solstice, both the 10° and 28° tilts produce nearly identical noon POA ($\sim$1020 W/m^2) because the sun passes almost overhead. In December, the 28° tilt delivers a visibly higher midday POA because the panel leans into the lower sun altitude.

A. Annual Tilt Sweep

Sweeping the tilt angle β from 0° to 55° and integrating POA over all 365 days of the year at NIT Trichy's latitude reproduces a shallow annual-yield curve typical of equatorial sites (Fig. 4). The computed optimum sits near $\beta \approx 10°$ for a south-facing panel, close to the site latitude. The curve is flat enough that any tilt between roughly 0° and 25° stays within a couple of percent of the peak, which is consistent with installed-PV practice where horizontal or slightly tilted rails are preferred near the tropics.

V. Single-Diode PV Electrical Model

The electrical model is implemented as an explicit $I(V)$ Newton solver around the standard single-diode form,

$$I = I_{\mathrm{ph}} - I_0\left[\exp\left(\frac{V + IR_s}{nN_sV_t}\right) - 1\right] - \frac{V + IR_s}{R_{\mathrm{sh}}}, \quad (11)$$

with thermal voltage $V_t = k_B(T+273.15)/q$. Temperature and irradiance are applied as coefficient updates on I_{sc} and V_{oc}:

$$I_{\mathrm{ph}}(G,T) = I_{\mathrm{sc,ref}} \cdot (G/1000) \cdot [1 + \alpha_{\mathrm{Isc}}(T-25)], \quad (12)$$

$$V_{\mathrm{oc}}(G,T) = V_{\mathrm{oc,ref}} \cdot [1 + \beta_{\mathrm{Voc}}(T-25)] \cdot \left[1 + 0.06 \ln\tfrac{G}{1000}\right], \quad (13)$$

and the saturation current $I_0 = I_{\mathrm{ph}}/(e^{V_{\mathrm{oc}}/V_t} - 1)$ is derived so that the computed open-circuit voltage is self-consistent. Table I lists the default 80 W educational module parameters used throughout this paper. Two additional presets (200 W mono-crystalline, 50 W poly-crystalline) ship with the app for comparison.

TABLE I
Default PV module reference parameters

Parameter	Value	Parameter	Value
$I_{\mathrm{sc,ref}}$	5.0 A	α_{Isc}	$+5\times10^{-4}$/°C
$V_{\mathrm{oc,ref}}$	22.0 V	β_{Voc}	-3.2×10^{-3}/°C
$I_{\mathrm{mp,ref}}$	4.55 A	γ_{Pmax}	-4.5×10^{-3}/°C
$V_{\mathrm{mp,ref}}$	17.6 V	Cells in series N_s	36
R_s	0.15 Ω	R_{sh}	250 Ω
Ideality n	1.3	NOCT	45 °C
Area	0.65 m^2	η_{ref}	15.5%

A. Irradiance Sweep

Figure 5 shows the computed I–V and P–V curves at $T_{\mathrm{cell}} = 25$ °C and four irradiance levels from 250 to 1000 W/m^2. The current scales almost linearly with irradiance as expected, while the voltage shifts only weakly. The open-circuit voltage changes by about 0.75 V between 1000 W/m^2 and 250 W/m^2, captured by the logarithmic irradiance correction term. The maximum power falls from 76.6 W to 18.6 W between the two extremes.

B. Temperature Sweep

Figure 6 holds the irradiance at 1000 W/m^2 and sweeps the cell temperature from 15 °C to 65 °C. As expected, V_{oc} drops noticeably with temperature (approximately -70 mV/°C from the β_{Voc} coefficient), I_{sc} rises slightly, and the net effect is a $P_{\max}$ reduction consistent with $\gamma_{\mathrm{Pmax}} \approx -0.45$%/°C. This is precisely the behaviour that a student is told to expect but rarely has a chance to watch interactively.

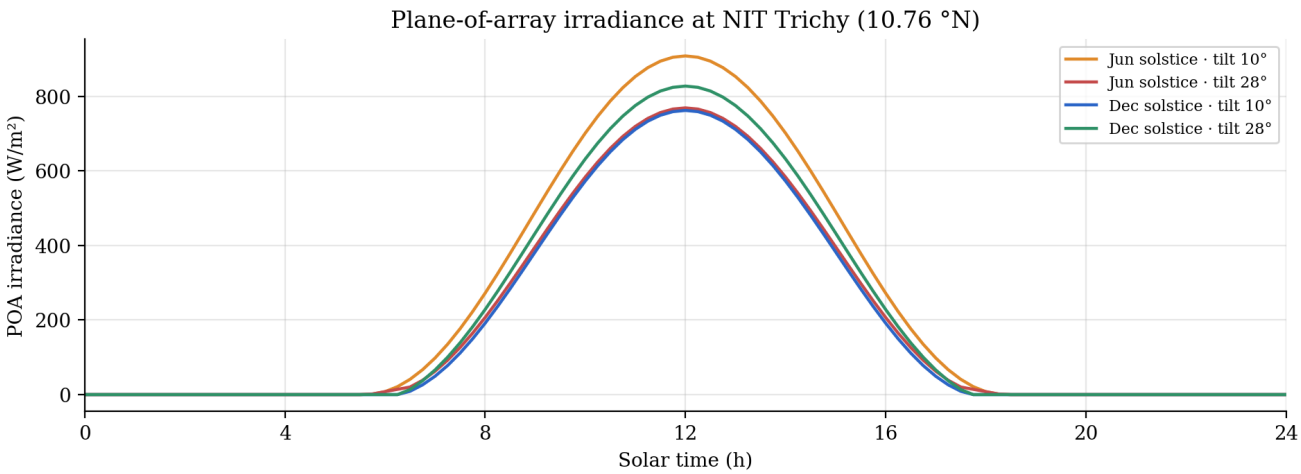

Fig. 3. Simulated daily POA irradiance at NIT Trichy for two tilt angles on June and December solstices.

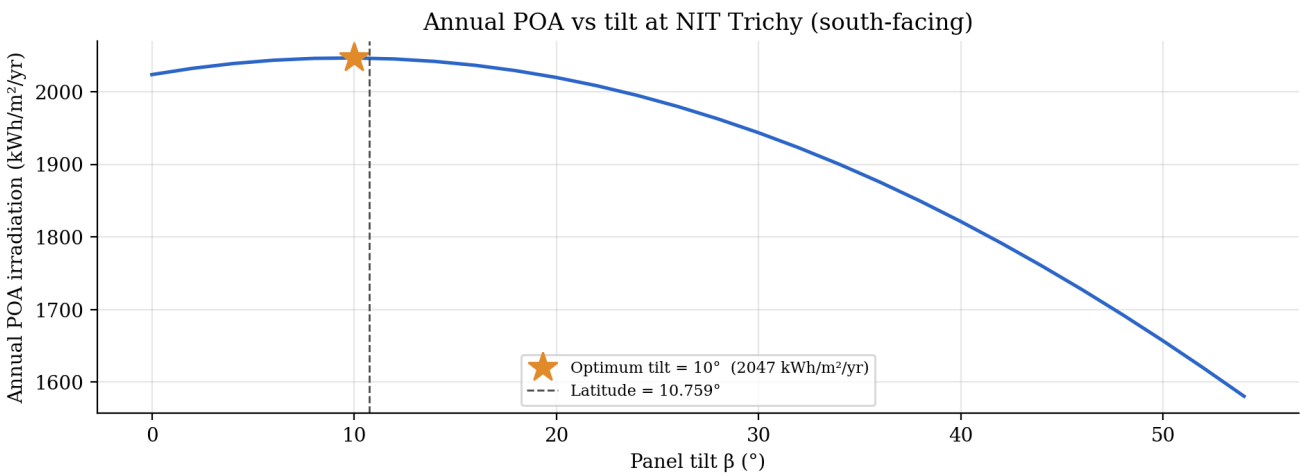

Fig. 4. Annual POA irradiation as a function of panel tilt at NIT Trichy, south-facing ($\gamma = 0$).

VI. Partial Shading and Bypass Diode

The module is treated as three series-connected substrings, each with its own effective irradiance multiplier. At each current level, the string voltage is the sum of the three substring voltages. When the bypass diode model is disabled, a strongly shaded substring clamps the string current; when it is enabled, the shaded substring is allowed to sit at approximately $-0.7\,\mathrm{V}$, letting the remaining substrings continue to deliver current.

Figure 7 compares the two cases for a configuration in which substring 2 receives only 20% of the irradiance of substrings 1 and 3. Without a bypass diode, the combined $P_{\max}$ collapses to a small fraction of its unshaded value. With the bypass diode enabled, the string recovers the bulk of the available power; the P–V curve develops the characteristic multi-peak shape that the lecture notes describe as the motivation for multi-step MPPT schemes. The test scenario Partial Shading ships with the app to reproduce this comparison live.

VII. Maximum Power Point Tracking

Two MPPT algorithms are implemented as pure functions stepped by an setInterval loop:

- Perturb & Observe (P&O): apply a small ΔV, compare current power to the previous sample, keep the same direction if power rose, reverse otherwise.
- Incremental Conductance (IncCond): compute dI/dV numerically and compare with $-I/V$. The MPP is reached when $dI/dV + I/V = 0$.

Figure 8 shows a P&O run on a P–V curve corresponding to $900\,\mathrm{W/m^2}$ at $35\,^\circ\mathrm{C}$, with $\Delta V = 0.25\,\mathrm{V}$ and a starting voltage of $5\,\mathrm{V}$. The controller climbs to within 1% of $P_{\max}$ in roughly sixty iterations and then oscillates within the perturbation band, which is the expected steady-state behaviour. The app also exposes an Incremental Conductance mode for qualitative comparison.

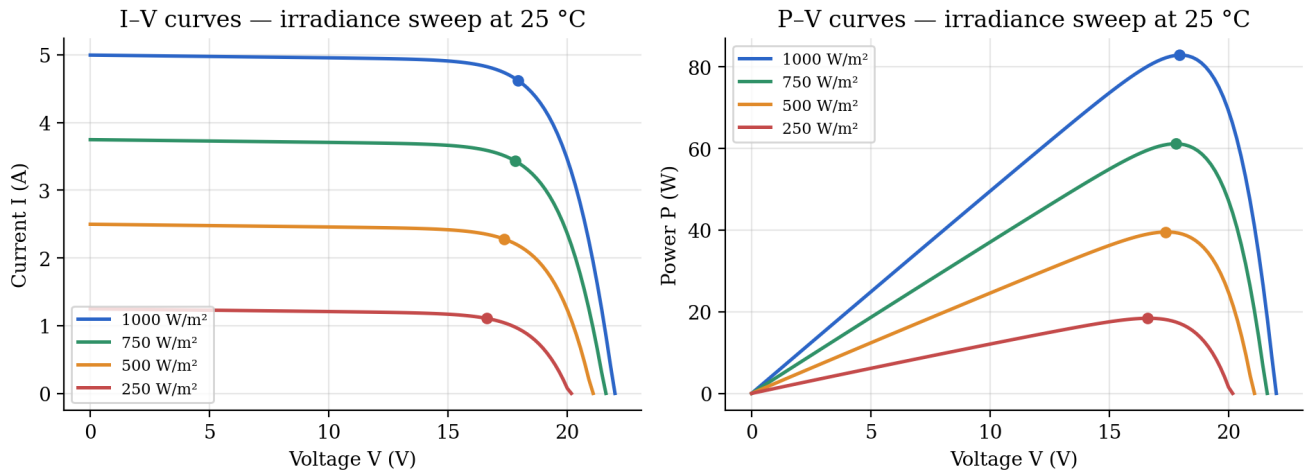

Fig. 5. Simulated I–V and P–V curves at $T_{\text{cell}} = 25\,°\text{C}$ for four irradiance levels. Circles mark the maximum power point.

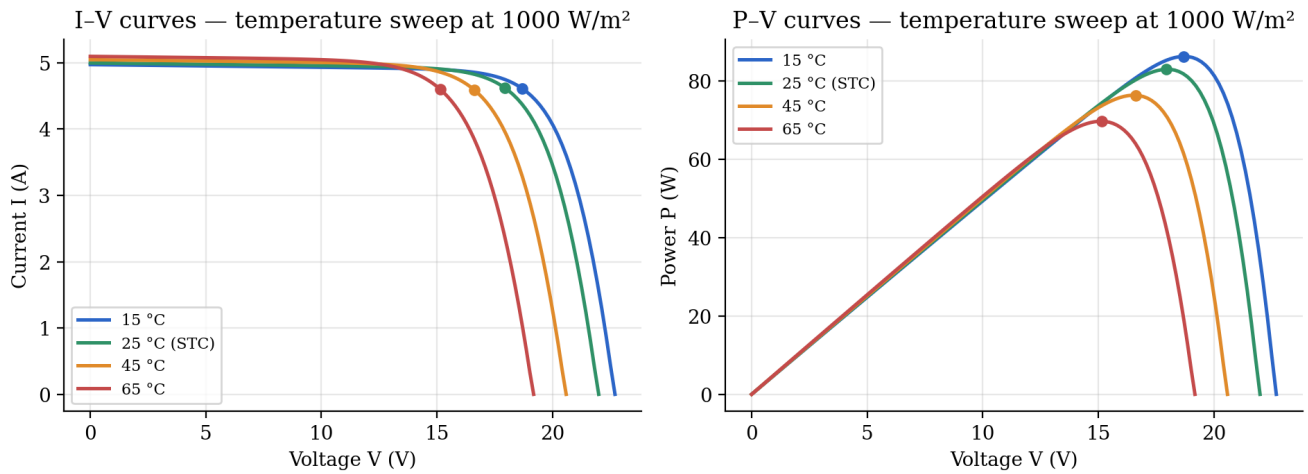

Fig. 6. Simulated I–V and P–V curves at $1000\,\text{W/m}^2$ for four cell temperatures.

VIII. NIT Tiruchirappalli Campus View

A distinguishing feature of this project is the NIT Trichy tab, which combines three real data sources to ground the abstract simulation in a concrete site:

1) 308 OpenStreetMap building footprints fetched via Overpass API for the bounding box [10.7500°N, 10.7870°N] × [78.8030°E, 78.8275°E], including named landmarks such as Sapphire Hostel, Ruby, Topaz, Jasper Hostel, A Mess, and the Golden Jubilee Building.
2) Extras: 784 road segments, 4 water features, 8 man-made water towers, 47 amenity elements, barrier lines, and place labels parsed from a second Overpass query.
3) Esri World Imagery aerial tile exported for the same bounding box at 2048×3091 pixels (aspect-ratio matched to the geographic bbox) and used as the ground texture.

Each building footprint is extruded to a height derived from the height or building:levels OSM tags where available, otherwise estimated by building type (12 m for hostel/residential, 9 m for university/academic, 3.5 m for shed/utility). Rooftops and walls receive distinct procedural window patterns through a custom shader grafted into the standard PBR material via onBeforeCompile. Water towers are rendered as 3D cylindrical tanks on legs. A directional light positioned at the sun vector casts shadow-mapped shadows onto both the satellite ground plane and the other buildings, so the same solar-geometry module that drives the panel scene is also driving the campus-scale shadow study.

A. Rooftop PV Potential

Table II and Fig. 9 summarise the rooftop PV potential computed from the OSM footprints. Assuming a 60% usable roof fraction (typical for academic campuses with HVAC, water tanks and walkways on roofs) and $180\,\text{W/m}^2$

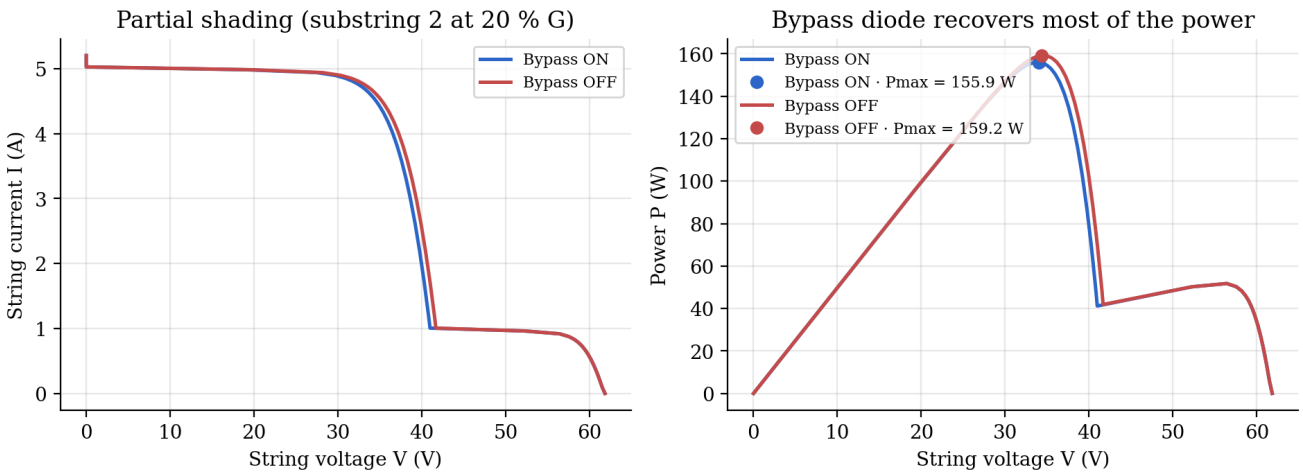

Fig. 7. String-level I–V and P–V curves with substring 2 at 20% irradiance. The bypass diode recovers most of the mismatch-induced power loss.

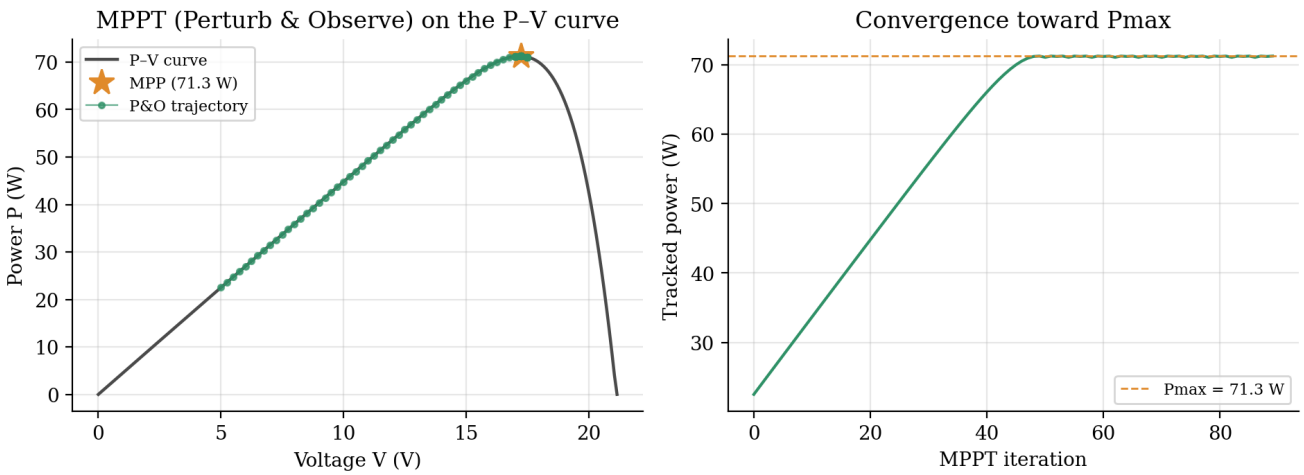

Fig. 8. Perturb & Observe MPPT homing in on the maximum power point (left) and the resulting tracked-power trajectory (right).

of installable PV, the aggregate peak capacity exceeds 20 MW.

TABLE II
NIT Trichy rooftop PV potential (OSM footprints)

Quantity	Value
Number of building footprints	308
Total gross footprint area	13.6 ha
Assumed usable fraction	60%
Total usable rooftop area	8.1 ha
Installed PV density	180 W/m^2
Estimated aggregate peak capacity	$\approx$ 14.6 MW$_p$
Expected annual yield (5.55 kWh/m^2/d, $\eta_{sys} = 80\%$)	$\approx$ 13.2 GWh/yr

IX. Site Climate from NASA POWER

The solar-resource sliders are seeded, at one click, from the NASA POWER RE community API [3] for any user-picked lat/lon. The repository ships pre-fetched datasets for NIT Trichy: a multi-decade monthly climatology and 365 days of 2023 daily data. Figure 10 plots the monthly climatology. The annual-mean values are GHI = 5.55 kWh/m^2/day, DNI = 3.64 kWh/m^2/day, DHI = 2.48 kWh/m^2/day, and T_{2m} = 27.6 °C. The monthly profile is strongly tropical: peak GHI in March–April (6.4–6.5 kWh/m^2/day), a minor monsoon dip from June through September, and a second trough in November–December (4.3 kWh/m^2/day) coinciding with the northeast monsoon.

Figure 11 shows the 2023 daily series with 7-day moving averages overlaid. Day-to-day variability is substantial during the monsoon months, where GHI can drop below 3 kWh/m^2/day; this is a first-order reason students should expect real-world annual yield to fall well below the naive rated power × 365 × 24 number.

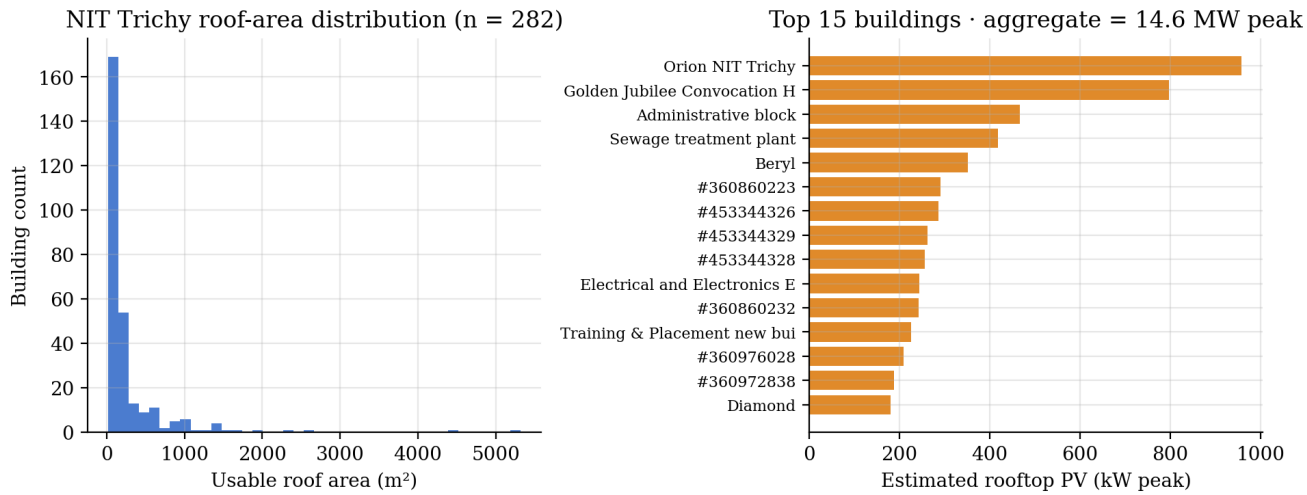

Fig. 9. Distribution of usable roof areas at NIT Trichy (left) and the top-15 named buildings ranked by estimated rooftop PV capacity (right).

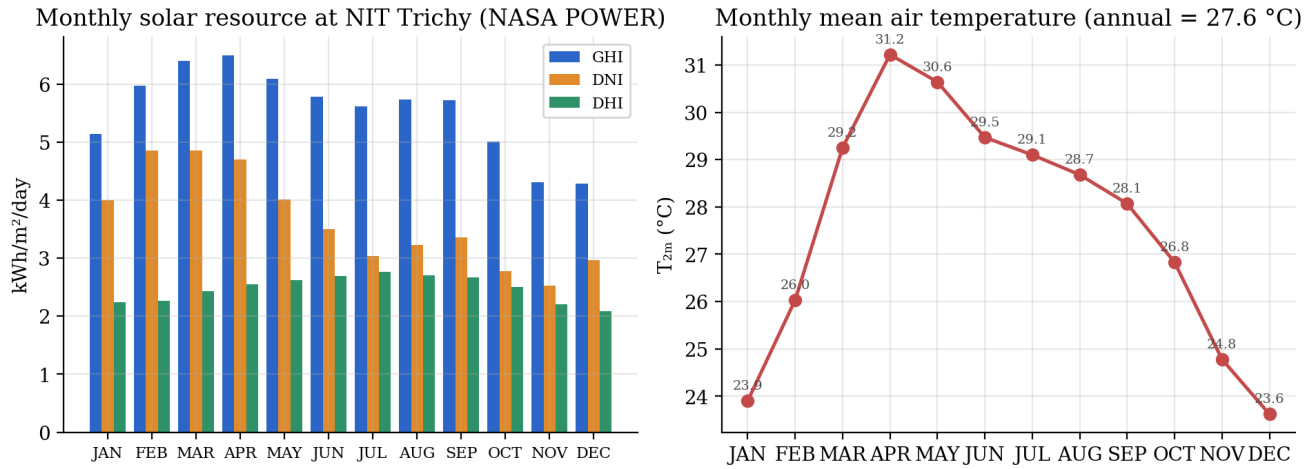

Fig. 10. NASA POWER monthly climatology at NIT Trichy (10.759 °N, 78.817 °E).

X. Acceptance Test Results

Table III lists the acceptance criteria stated in the original project specification and the observed behaviour of the final build. All 13 criteria pass in the shipped application.

Beyond the minimum set, the project delivers three stretch features called out in the original spec: multiple panel presets, an annual tilt sweep, and a comparison mode for two operating conditions (implemented as the bypass-on / bypass-off view). Two further features not in the original spec were added because the course notes motivated them: a NASA POWER climate importer and the NIT Trichy campus view.

XI. Limitations and Future Work

Three limitations should be stated explicitly.

1) The POA model is the isotropic-sky approximation. The app does not yet implement anisotropic diffuse models (Hay-Davies, Perez). For low-altitude sunrise/sunset cases with significant circumsolar diffuse, this will under-predict POA.
2) The cell-temperature estimate is the NOCT linearisation $T_{\text{cell}} = T_{\text{amb}} + (\text{NOCT} - 20)/800 \cdot G_{\text{POA}}$. A full thermal model would require wind-speed and mounting-type inputs that are out of scope for a v1 educational tool.
3) OSM building footprints at NIT Trichy are volunteer-traced and occasionally off by 5–30 m relative to satellite imagery. The satellite ground is aspect-correct and pixel-aligned to the same local metre coordinate frame, but the 3D footprints themselves inherit OSM's geocoding accuracy. Manual alignment via the in-browser JOSM editor would fix this for any future production deployment.

Natural extensions include: (i) an annual simulation

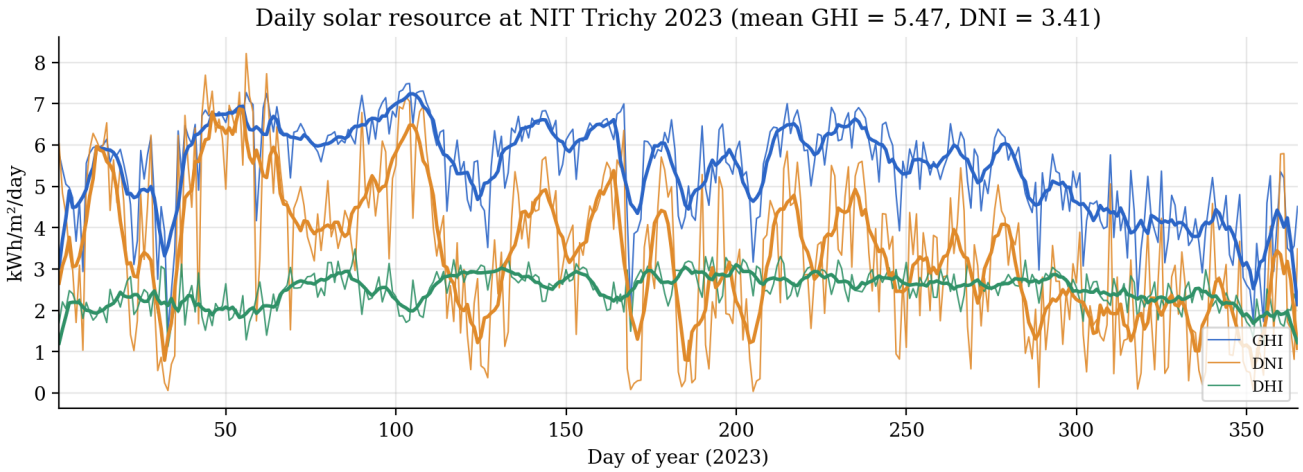

Fig. 11. Daily GHI, DNI and DHI at NIT Trichy for all 365 days of 2023, with 7-day moving averages.

TABLE III
Acceptance criteria from the original project specification

#	Criterion	Verified behaviour
1	User opens browser page and sees a 3D solar panel scene	Panel Scene tab renders at 60 fps on a 2021 MBP
2	Changing date/time/latitude changes sun direction and incidence angle	Driven by computeSolarAngles(); live metric update
3	Changing tilt/azimuth changes incident irradiance on the panel	Panel colour + POA metric update on every slider drag
4	Beam, diffuse, reflected and total irradiance displayed	Metrics panel, live numeric readout
5	Live I–V and P–V curves rendered	PV Characteristics tab (Recharts)
6	$I_{sc}, V_{oc}, I_{mp}, V_{mp}, P_{max}$, FF displayed	Metrics panel (6 tiles)
7	Temperature rise visibly lowers V_{oc} and P_{max}	Confirmed by Fig. 6
8	Irradiance rise raises current and power	Confirmed by Fig. 5
9	Partial shading reduces output and changes the curve	Shading & Diodes tab, substring sliders
10	Bypass diode ON/OFF visibly changes shaded case	Side-by-side P_{max} comparison (Fig. 7)
11	MPPT mode converges toward MPP	Green-dot tracker on P–V chart (Fig. 8)
12	Daily irradiation / daily energy estimate displayed	Metrics panel and daily-irradiance chart
13	UI is understandable without reading source code	Theory tab, tooltips, preset buttons

mode across all 365 days of NASA POWER data; (ii) a battery-plus-load demonstration module tying the MPPT output to a simple energy balance; (iii) drone photogrammetry of the NIT Trichy campus to replace the OSM footprints with a photoreal mesh; (iv) CSV export of every chart for downstream analysis.

XII. Conclusion

This paper has described a course-aligned, fully browser-based 3D solar PV simulator that unifies solar geometry, POA irradiance, single-diode PV modelling, partial shading with bypass diode behaviour, and MPPT animation in a single coherent tool. A dedicated campus-scale view grounds the abstract simulation in a real site — NIT Tiruchirappalli — using 308 OpenStreetMap building footprints rendered on an Esri satellite tile, with solar angles driven by the same physics modules that drive the panel-scale scene. Computed results reproduce the expected textbook behaviour across the full parameter space, and a rooftop PV potential study estimates approximately 14.6 MW_p of usable roof-mounted PV capacity on campus. The complete source code, bundled OSM/NASA datasets, and this report are open-sourced at https://github.com/KaranSinghBisht/solar-pv-simulator.

Acknowledgment

The authors thank the NIT Tiruchirappalli Department of Electrical and Electronics Engineering for the course framing, the OpenStreetMap community for the NIT Trichy building and road data, NASA's POWER project for the solar-resource time series, and the Esri World Imagery service for the campus aerial texture.

References

[1] W. F. Holmgren, C. W. Hansen, and M. A. Mikofski, "pvlib python: A python package for modeling solar energy systems," Journal of Open Source Software, vol. 3, no. 29, p. 884, 2018.
[2] OpenPV Initiative, "simshady: Simulating shadows for PV potential analysis with 3D data in the browser," GitHub repository, 2024. [Online]. Available: https://github.com/open-pv/simshady
[3] NASA Langley Research Center, "POWER: Prediction Of Worldwide Energy Resources," NASA, 2024. [Online]. Available: https://power.larc.nasa.gov
[4] R. Olbricht, "Overpass API," OpenStreetMap, 2024. [Online]. Available: https://overpass-api.de
[5] Esri, "World Imagery," ArcGIS Online, 2024. [Online]. Available: https://www.arcgis.com/home/item.html?id=10df2279f9684e4a9f6a7f08febac2a9

[6] J. A. Duffie and W. A. Beckman, Solar Engineering of Thermal Processes, 4th ed. Hoboken, NJ, USA: Wiley, 2013.
[7] G. M. Masters, Renewable and Efficient Electric Power Systems, 2nd ed. Hoboken, NJ, USA: Wiley, 2013.
[8] N. Femia, G. Petrone, G. Spagnuolo, and M. Vitelli, "Optimization of perturb and observe maximum power point tracking method," IEEE Trans. Power Electron., vol. 20, no. 4, pp. 963–973, Jul. 2005.